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Product : T ZORAXIN SUSPENSION 200MG/5ML [FRONT] SAP No. : Date : 15 November 2024		☐ Unit Box ☐ Outer Box ☐ Label		☒ Leaflet ☐ Fix-A-Form ☐ Shipper Carton	
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ZORAXIN SUSPENSION 200MG/5ML

DESCRIPTION:
ZORAXIN SUSPENSION 200MG / 5ML is a yellow suspension.

COMPOSITION:
Each 5 ml suspension contains Acyclovir 200 mg.
Preservatives:
Methyl Paraben 0.1% w/v
Propyl Paraben 0.03% w/v

PHARMACODYNAMICS:
Acyclovir is a synthesis purine nucleoside analogue with in vitro and in vivo inhibitory activity against human herpes viruses, including herpes simplex virus (HSV) types 1 and 2. Varicella zoster virus (VZV), Epstein Barr virus (EBV) and Cytomegalovirus (CMV).
Acyclovir has the greatest antiviral activity against HSV-1, followed (in decreasing order or potency) by HSV-2, VZV, EBV and CMV. The inhibitory activity of acyclovir for HSV-1, HSV-2, VZV, EBV and CMV is highly selective. The enzyme thymidine kinase (TK) of normal, non-infected cells does not use acyclovir effectively as a substrate, hence toxicity to mammalian host cells is low; however, TKs encoded by HSV, VZV, and EBV convert acyclovir to acyclovir monophosphate, a nucleoside analogue which is further converted to the diphosphate and finally to the triphosphate by cellular enzymes. Acyclovir triphosphate interferes with the viral DNA polymerase and inhibits viral DNA replication with resultant chain termination following its incorporation into the viral DNA. Prolonged or repeated courses of acyclovir in severely immune-compromised individuals may result in the selection of virus strains with reduced sensitivity, which may not respond to continued acyclovir treatment. Isolates with reduced sensitivity have been relatively deficient in viral TK, however, strains with altered viral TK or DNA polymerase have also been reported. Exposure of HSV isolates to acyclovir can also lead to the emergence of less sensitive strains. The relationship between the in-vitro determined sensitivity of HSV isolates and clinical response to acyclovir therapy is not clear.

PHARMACOKINETICS:
Absorption: Bioavailability 20% (range, 15 to 30%) poorly absorbed from the gastrointestinal tract. Not significantly affected by food.
Distribution: Widely distributed to tissues and body fluids, including brain, kidneys, lungs, liver, aqueous humor, tears, intestines, muscle, spleen, breast milk, uterus, vaginal mucosa, vaginal secretions, semen, amniotic fluid, cerebrospinal fluid (CSF), and herpetic vesicular fluid. Highest concentrations are found in kidneys, liver, and intestines. CSF concentrations are approximately 50% of plasma concentrations. It also crosses the placenta
Protein binding: Low (9 to 33%).
Biotransformation: Hepatic; only major metabolite found in urine is 9-carboxymethoxymethylguanine, which accounts for approximately 9 to 14% of the dose. This metabolite has no known antiviral activity.
Half Life: Adult: 3.3 hours.
Time to peak serum concentrations: 1.7 hours.
Mean peak serum concentration (steady state):
200mg every 4 hours – 0.6mcg/mL (2.5 micromoles/L).
400mg every 4 hours – 1.2mcg/mL (5.3 micromoles/L).
800mg every 4 hours – 1.6mcg/mL (6.9 micromoles/L).
Elimination:
Renal: Approximately 14% of total dose excreted unchanged in urine.
Fecal: Insignificant amount (<2%).
Lungs: Trace amount in exhaled CO2.
Dialysis:
Haemodialysis – a single 6 hours period of haemodialysis reduces plasma acyclovir concentrations by approximately 60%.
Peritoneal Dialysis – Peritoneal dialysis does not substantially alter acyclovir clearance.

INDICATION:
Herpes simplex virus infections of the skin and mucous membranes, including initial and recurrent genital herpes. Suppression (prevention of recurrences) of recurrent herpes simplex infections, in immunocompetent patients. Prophylaxis of herpes simplex infections in immunocompromised patients. Herpes zoster (shingles) infections. Early treatment of shingles with acyclovir has a beneficial effect on pain and can reduce the incidence of post-herpetic neuralgia (zoster-associated pain).

RECOMMENDED DOSAGE:
Dosage of Herpes Simplex in Adults: 200 mg acyclovir should be taken 5 times daily at approximately 4-hourly intervals omitting the night-time dose. Treatment should continue for 5 days, but in severe initial infections may have to be extended. In severely immune-compromised cases (e.g. after bone marrow transplantation) or in cases with impaired absorption from the gut, the dose can be doubled to 400 mg. Dosing should begin as early as possible after the start of an infection; for recurrent episodes, this should preferably be during the prodromal period or when lesions first appear.
Suppression of herpes simplex in adults: In immune-competent cases, 200 mg acyclovir should be taken 4 times daily at approximately 6-hourly intervals. Many cases may be conveniently managed on a regimen of 400 mg acyclovir taken twice daily at approximately 12-hourly intervals. Dosage titration down to 200 mg acyclovir taken 3 times daily at approximately 8-hourly intervals or even twice daily at approximately 12-hourly intervals, may prove effective. Some cases may experience breakthrough infections on total daily doses of 800 mg acyclovir. Therapy should be interrupted periodically at intervals of 6 – 12 months in order to observe possible changes in the natural history of the disease.
Prophylaxis of herpes simplex in adults: In immune-compromised cases, 200 mg acyclovir should be taken 4 times daily at approximately 6-hourly intervals.
In severe immune-compromised cases (e.g. after bone marrow transplant) or in cases with impaired absorption from the gut, the dose can be doubled to 400 mg. The duration of prophylactic administration is determined by the duration of the period at risk.
Dosage of Herpes Zoster in adults: 800 mg 5 times daily at approximately four-hourly intervals, omitting the night-time dose. Therapy should commence as early as possible after the onset of rash but definitely within 72 hours of the appearance of the rash. Treatment should be continued for 7 days. For herpes zoster ophthalmicus, the recommended duration of therapy is 7 to 10 days. Attention should be given to maintaining adequate hydration in elderly patients.
Dosage in children: Herpes simplex infections in the immune-compromised children over the age of 2 years should be given adult dosages and children below the age of 2 years with the weight of >5kg should be given ½ the adult dose. No specific data are available on the suppression of herpes simplex infections or the treatment of herpes zoster infections in immune-competent children.
Use in the elderly: In the elderly, total acyclovir body clearance declines in parallel with creatinine clearance. Adequate hydration of elderly cases taking high oral doses of acyclovir should be maintained. Special attention should be given to dosage reduction in elderly cases with impaired renal function.
Renal impairment: For patients with severe renal impairment (creatinine clearance < 10 ml/min) an adjustment of dosage to 200 mg twice daily at approximately 12-hourly intervals is recommended. In herpes zoster infections, it is recommended to adjust the dosage to 800 mg twice daily at approximately 12-hourly intervals for cases with severe renal impairment (creatinine clearance < 10 ml/min), and to 800 mg 3 times daily at intervals of approximately 8 hrs for cases with moderate renal impairment (creatinine clearance in the range 10-25 mL/min).

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ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS: Patients known to be hypersensitive to acyclovir and propylene glycol.

WARNINGS & PRECAUTIONS:

The dosage should be adjusted in patients with severe renal impairment (see Dosage and Administration). It is important to maintain adequate hydration in elderly patients taking high doses of Acyclovir for the treatment of Herpes Zoster. The potential for the development of resistant HSV strains in patients treated with acyclovir should be borne in mind. As acyclovir has been associated with reversible encephalopathic changes, it should be used with caution in patients with underlying neurological abnormalities, significant hypoxia or serious renal, hepatic or electrolyte abnormalities. It should also be used with caution in patients who have manifested neurological reactions to cytotoxic drugs or are receiving concomitantly interferon or intrathecal methotrexate. Animal studies indicate that at high doses acyclovir is cytotoxic.

INTERACTION WITH OTHER MEDICAMENTS:

Probenecid increases the mean-half life and area under the plasma concentration curve of systemically-administered acyclovir. Other drugs affecting renal physiology could potentially influence the pharmacokinetics of acyclovir. However, no other drug interactions with acyclovir have been identified.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy:

Animal studies show that acyclovir crosses the placenta readily. There have been no adequate and well controlled studies concerning the safety of acyclovir in pregnant women. It should not be used during pregnancy unless the benefits to the patient clearly outweigh the potential risks to the foetus. If suppressive therapy is used in the perinatal period it should not be assumed that viral shedding has ceased, or that the risk to foetus/neonate has decreased. Pregnancy should be managed according to considerations normally applicable to patients with genital herpes.

Use in Lactation:

Limited human data show that acyclovir does pass into breast milk. Acyclovir should only be administered to nursing mothers if the benefits to the mother outweigh the potential risks to the baby.

SIDE EFFECTS:

Acyclovir appears to be generally very well tolerated. Adverse effects are usually mild. However the following have been noted:

Short Term Administration for Treatment for Genital Herpes: Nausea and/or vomiting and headache were the most frequent adverse effects. Less frequent (<1%) reactions included diarrhoea, dizziness, anorexia, fatigue, oedema, skin rashes, leg pain, inguinal adenopathy, medication taste and sore throat. Occasional changes in liver enzymes and changes in haematological parameters were also noted.

Long Term Suppressive Therapy for Genital Herpes: Nausea and/or vomiting, headache, diarrhoea, vertigo and arthralgia were the most frequent adverse effects. Less frequent adverse effects included skin rash, insomnia, fatigue, fever, palpitation, sore throat, superficial thrombophlebitis, muscle cramps, paronychia, menstrual abnormalities, lymphadenopathy, irritability, accelerated hair loss, depression and occasional increases in liver enzymes.

Treatment of Herpes Zoster: The most commonly reported adverse effect was gastro-intestinal disturbance. Other reports included aching, chest pain, confusion, constipation, diarrhoea, giddiness, hallucinations, headache, insomnia, nausea, rash, shaking, taste disturbance, tremor, vertigo and malaise, vomiting and mental status alteration.

Other : Reversible neurological reactions, notably dizziness, confusional states, hallucinations, somnolence and convulsions, have occasionally been reported, usually in patients with renal impairment in whom the dosage was in excess of that recommended or with other predisposing factors.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: Acyclovir is only partly absorbed in the gastrointestinal tract. It is unlikely that serious toxic effects would occur if a dose of up to 5 g were taken on a single occasion. No data are available on the consequences of the ingestion of higher doses. Ingestion of doses of acyclovir in excess of 5 g warrants close observation of the patient.

Treatment: Adequate hydration to prevent precipitation of acyclovir in the renal tubules. Haemodialysis to aid in the removal of acyclovir from the blood, especially in patients with acute renal failure and anuria.

STORAGE CONDITIONS:

Store below 30°C. Protect from light.
KEEP OUT OF REACH OF CHILDREN. JAUHKAN DARIPADA KANAK-KANAK.
Stability after opening: Store below 30°C at room temperature.
Please store in ambient temperature for not more than 4 weeks.
Discard if physical changes observed.

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Pack in 120 ml in in PET amber bottle.

PRODUCT REGISTRATION HOLDER:

Duopharma Marketing Sdn. Bhd.
Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13,
Bangi Industrial Estate, 43650 Bandar Baru Bangi,
Selangor, Malaysia.

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Revision Date:
November 2024

